Course Name: Vocational Skill Enhancement Course

Course Code: ET24305

An Activity Report on

Implementation of a D Flip-Flop (IC 74HC74) with an IC 555 Clock Circuit

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Abstract

This project demonstrates a clocked D flip-flop circuit using an NE555 timer and a 74HC74 dual D-type flip-flop integrated circuit. The NE555 is configured in astable mode to generate continuous clock pulses. These pulses are applied to the clock input of the 74HC74. By connecting the D input of the first flip-flop to its complementary output, the circuit toggles state on each rising clock edge. Two LEDs with current-limiting resistors indicate the complementary output states. The report also describes a single-sided printed circuit board (PCB) layout concept, component selection, circuit operation, testing procedure, precautions, and possible applications. The PCB preview included is a planning illustration; a manufacturing design must be checked in PCB CAD software before fabrication.

1. Introduction

A D flip-flop is a basic sequential logic element used to store one bit of information. Unlike combinational circuits, its output changes in response to a clock event. The 555 timer is a widely used timing IC that can produce a periodic square wave when connected in astable mode. Combining the timer with a D flip-flop provides a simple demonstration of clocked storage and frequency division.

In this implementation, the NE555 supplies the clock signal and the 74HC74 performs the storage and output switching. The PCB arrangement is intended for a regulated 5 V DC supply and through-hole components, making it suitable for an electronics laboratory demonstration.

2. Aim

To implement a D flip-flop using an NE555 timer as a clock-pulse generator and a 74HC74 IC, and to observe the output states using LEDs on a single-sided PCB.

3. Objectives

- To study the astable operation of the NE555 timer.
- To generate periodic clock pulses.
- To understand the clocked operation of a D flip-flop.
- To observe complementary outputs Q and $\overline{Q}$ using LEDs.
- To prepare a single-sided PCB layout concept for the circuit.

4. Components Required

Component	Value / Type	Quantity
Timer IC	NE555, DIP-8	1
D flip-flop IC	74HC74, DIP-14	1
Voltage Regulator	LM7805	1
Timing resistor RA	3.9 kΩ, ¼ W	1

Timing resistor RB	68 kΩ, ¼ W	1
LED resistors	330 Ω, ¼ W	2
Timing capacitor C1	10 µF electrolytic	1
Control capacitor C2	10 nF ceramic	1
Decoupling capacitors	100 nF ceramic (recommended)	2
LEDs	Red and green	2
Input connector	2-pin terminal block	1
Switch	Push Button	1
Supply	Regulated +5 V DC	1
PCB	Single-sided FR-4, proposed 50 × 35 mm	1

5. Theory

5.1 NE555 Timer in Astable Mode

In astable mode, the NE555 repeatedly charges and discharges a timing capacitor through resistors RA and RB. This produces a continuous rectangular waveform at pin 3. For the conventional astable connection, the approximate timing equations are: $T_{HIGH} \approx 0.693(RA + RB)C$ and $T_{LOW} \approx 0.693(RB)C$. Therefore, $f \approx 1.44/[(RA + 2RB)C]$. With RA = 3.9 kΩ, RB = 68 kΩ and C = 10 µF, the ideal frequency is approximately 1.0 Hz; component tolerances and the capacitor's leakage can change the actual value.

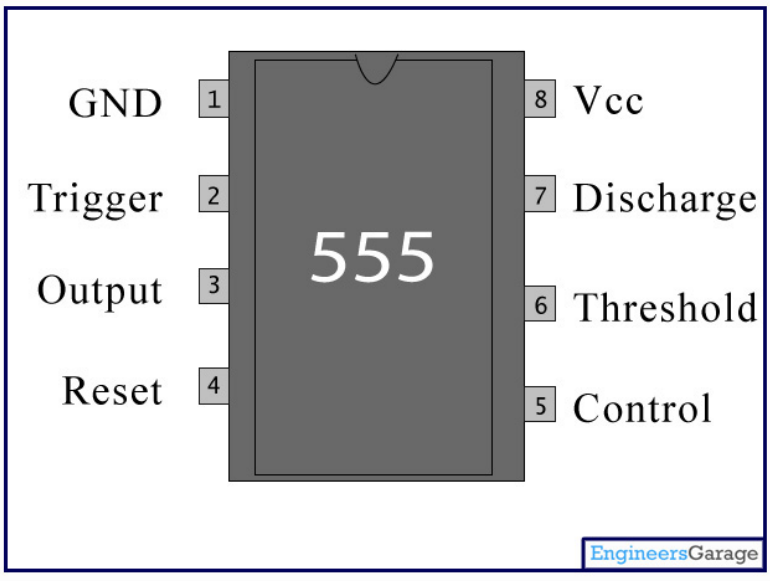


Fig 1: Pin Configuration of IC 555

5.2 D Flip-Flop

The 74HC74 is a dual positive-edge-triggered D flip-flop. On the rising edge of the clock, the logic level at D is copied to Q. The complementary output $\overline{Q}$ represents the inverse state. Between active clock edges, the stored state remains unchanged unless asynchronous preset or clear is activated.

Characteristic relation: Q(next) = D, sampled at the active rising clock edge.

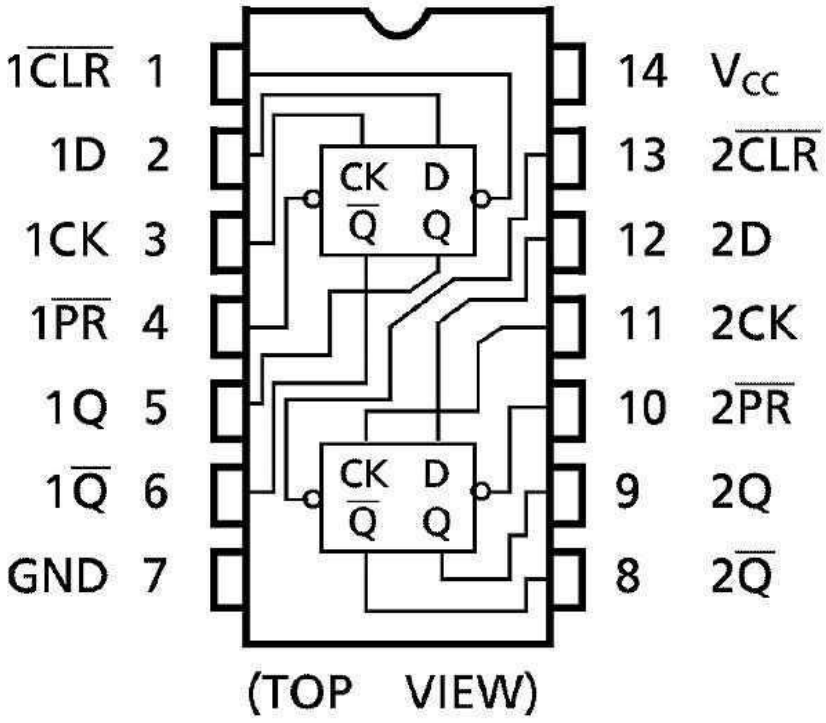


Fig 2:Pin Configuration of IC 7474

6. Circuit Connections

6.1 NE555 Connections

- Pin 1 (GND) → ground.
- Pin 8 (VCC) → +5 V.
- Pin 4 (RESET) → +5 V.
- Pins 2 (TRIG) and 6 (THRESH) → joined timing node.
- Pin 7 (DISCH) → junction of RA and RB.
- RA = 3.9 kΩ between +5 V and pin 7.

- RB = 68 kΩ between pin 7 and the pins 2/6 timing node.
- C1 = 10 µF from the timing node to ground; capacitor positive terminal to the timing node.
- Pin 5 (CTRL) → 10 nF capacitor to ground.
- Pin 3 (OUT) → 74HC74 pin 3 (CLK1).

6.2 74HC74 Connections

- Pin 14 (VCC) → +5 V; pin 7 (GND) → ground.
- Pin 3 (CLK1) → NE555 pin 3.
- Pin 2 (D1) → 1 k resistor → push button
- Pin 1 (/CLR1) and pin 4 (/PRE1) → +5 V for normal operation.
- Pin 5 (Q1) → 330 Ω resistor → LED1 anode; LED1 cathode → ground.
- Pin 6 ($\bar{Q}$1) → 330 Ω resistor → LED2 anode; LED2 cathode → ground.
- For the unused second flip-flop, tie pins 10 (/PRE2) and 13 (/CLR2) HIGH; tie pins 11 (D2) and 12 (CLK2) to a defined logic level such as ground. Leave pins 8 and 9 unconnected.
- Place a 100 nF decoupling capacitor close to each IC between its supply and ground pins.

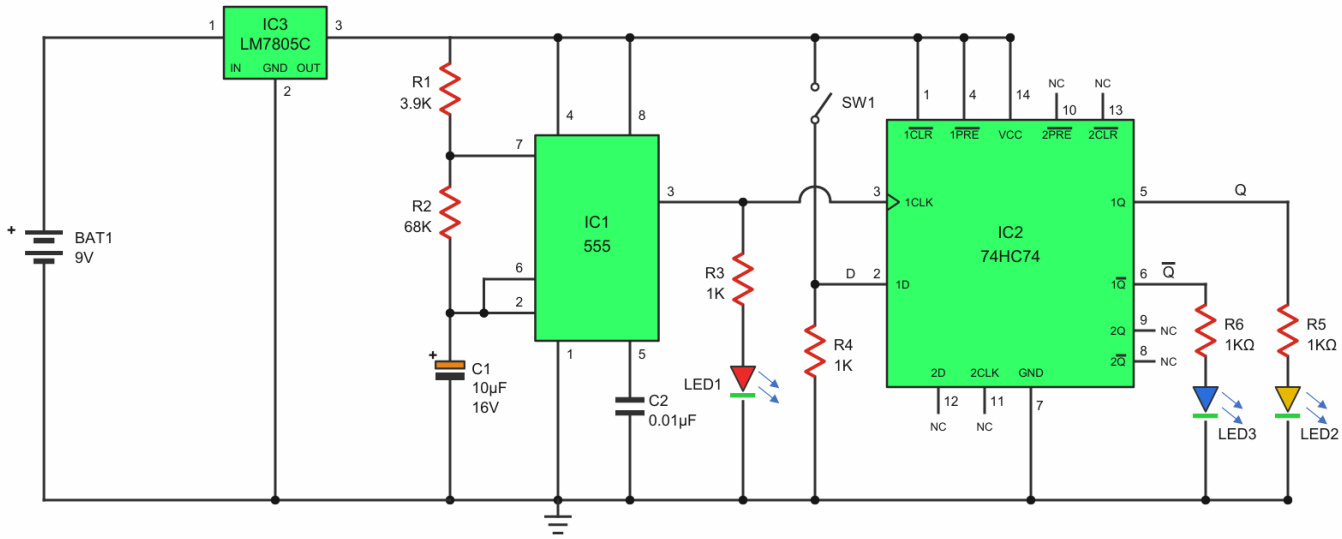


Fig 3: Circuit Diagram

7. Working Principle

The circuit consists of two main integrated circuits: the NE555 timer IC and the 74HC74 D flip-flop IC. The NE555 timer generates clock pulses, while the 74HC74 stores the data input and produces the output. A push button is used to provide logic 1 or logic 0 at the data input, and two LEDs indicate the output states.

Step-by-step working

1. Clock pulse generation using NE555

   The NE555 timer is connected in astable mode. The timing capacitor continuously charges and discharges through the timing resistors, producing a periodic square-wave signal at pin 3. This signal acts as the clock input for the D flip-flop.

2. Applying data input

   The push button is connected to the D input (pin 2) of the 74HC74 IC. When the button is pressed, the D input receives logic HIGH (1). When the button is released, the 10 kΩ pull-down resistor makes the D input logic LOW (0).

3. Operation of the D flip-flop

   The 74HC74 is a positive-edge-triggered D flip-flop. It reads the value at the D input only when the clock changes from LOW to HIGH (the rising edge). At that instant, the input value is transferred to output Q.

   1. If D = 0 at the rising clock edge, Q becomes 0 and $\bar{Q}$ becomes 1.
   2. If D = 1 at the rising clock edge, Q becomes 1 and $\bar{Q}$ becomes 0.
   3. Between rising clock edges, the output retains its previously stored state.

4. Output indication using LEDs

   The Q output (pin 5) and complementary output $\bar{Q}$ (pin 6) are connected to separate LEDs through current-limiting resistors. When Q is HIGH, LED 1 lights and LED 2 remain OFF. When Q is LOW, LED 1 remains OFF and LED 2 lights, assuming the LEDs are connected from each output through a resistor to ground.

5. Data storage

   The circuit stores the input value until the next active rising clock edge. Therefore, changing the push button between clock edges does not immediately change the output. The output updates only at the next rising clock edge.

8. Truth Table

Clock condition	D input	Q after event	$\bar{Q}$
Rising edge	0	0	1
Rising edge	1	1	0
No active edge	X	Previous state	Previous state

X means either logic 0 or logic 1. When D is connected to $\bar{Q}$, Q toggles at each rising edge.

9. PCB Layout and Design

The proposed PCB is single-sided and uses through-hole components to simplify assembly. The NE555 timing network is placed near the NE555 IC, while the 74HC74 and LED output components are grouped together. A two-pin terminal block is used for the +5 V and ground input. The suggested

outline is 50 mm × 35 mm with four mounting holes; the final dimensions and hole locations should be confirmed in PCB CAD software.

The following image is a visual concept for report presentation, not a dimensionally accurate copper artwork. It must not be sent directly to a PCB manufacturer. Before fabrication, create or verify the actual footprints, route all copper tracks on one copper layer, run electrical-rule and design-rule checks, and generate Gerber and drill files from the verified CAD design.

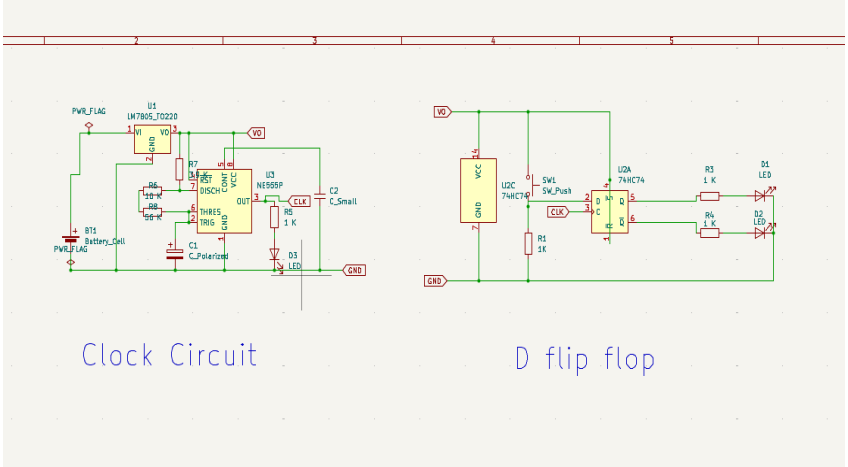
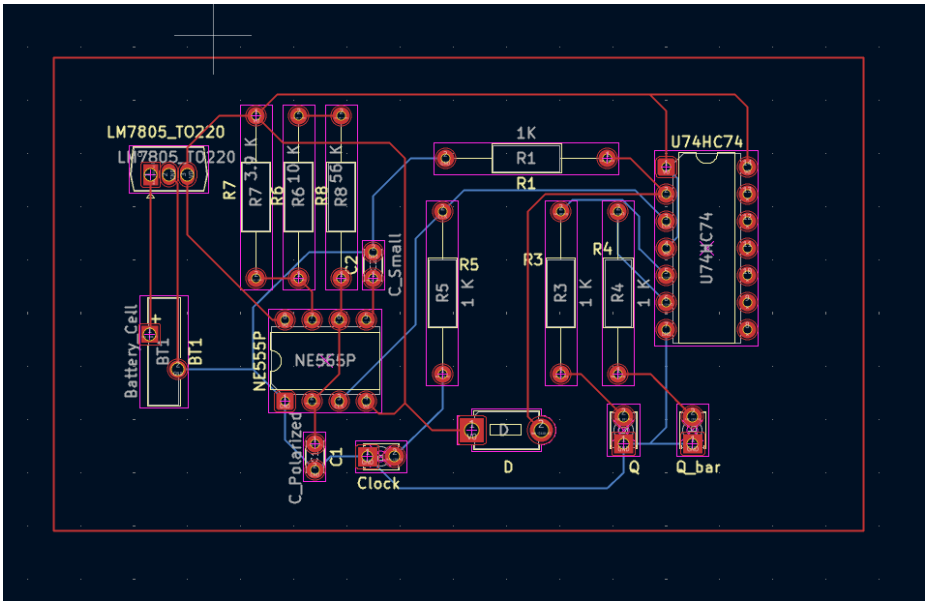


Fig 4: Schematic Design



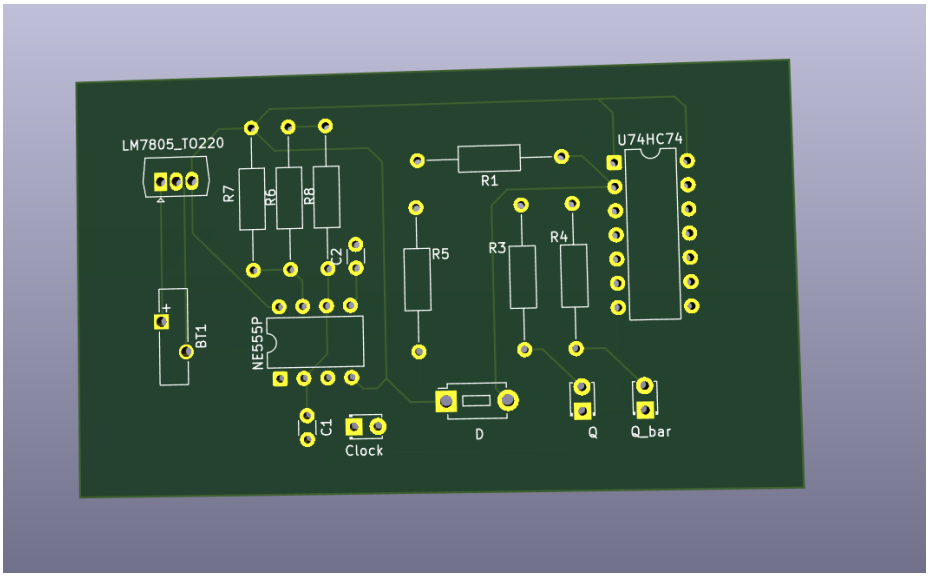


Fig 5 : PCB Layout

10. Procedure

1. Check the datasheets and identify pin 1 on both ICs.
2. Assemble the NE555 astable timing circuit on a breadboard first.
3. Verify the clock waveform at NE555 pin 3, if an oscilloscope or logic probe is available.
4. Connect the 74HC74 supply, clock, D feedback, preset, and clear pins as described.
5. Connect both LEDs with their 330 Ω current-limiting resistors.
6. Apply a regulated 5 V supply and observe the LEDs.
7. After confirming operation, transfer the circuit to the PCB layout and verify all connections before soldering.

11. Expected Observations

- The NE555 output produces a periodic clock waveform.
- The Q output changes state at each rising clock edge when D is connected to $\overline{Q}$.
- Q and $\overline{Q}$ show complementary logic levels.
- The LEDs indicate alternating output states, subject to LED orientation and wiring.

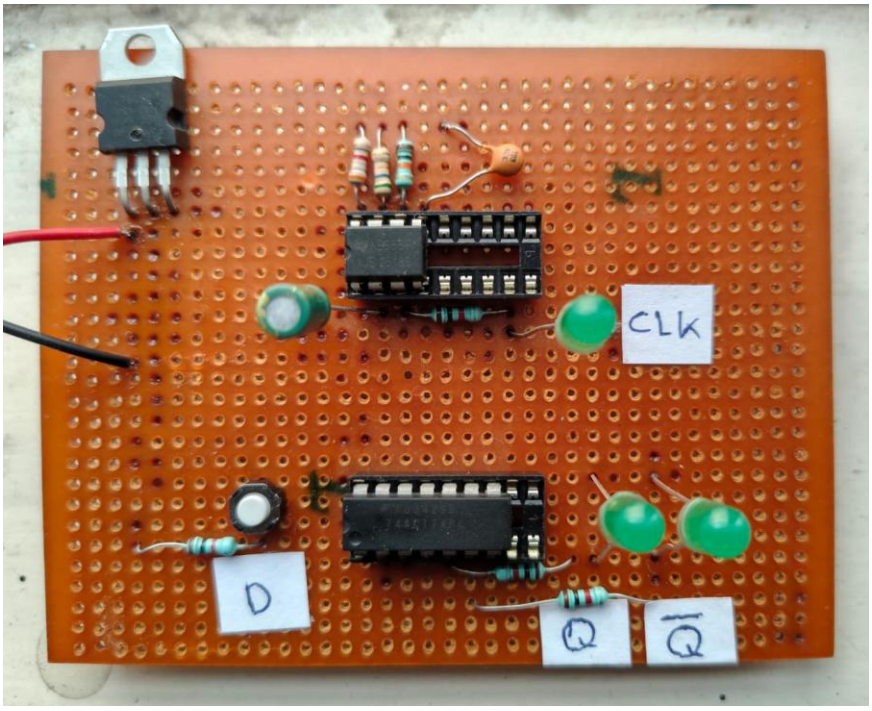


Fig 6: Final Circuit

12. Applications

- One-bit data storage and transfer.
- Registers and sequential logic circuits.
- Frequency division and simple counters when configured as a toggle flip-flop.
- Clocked digital demonstrations and laboratory experiments.

13. Precautions

- Use a regulated 5 V supply and common ground.
- Verify IC pin numbering and package orientation before powering the circuit.
- Do not leave asynchronous preset/clear pins or unused flip-flop inputs floating.
- Use current-limiting resistors with LEDs.
- Observe electrolytic capacitor polarity.
- Check for shorts and run PCB design-rule checks before manufacturing.

14. Conclusion

The project demonstrates a clocked D flip-flop using a 74HC74 IC driven by an NE555 astable timer. The flip-flop stores the input logic state on a rising clock edge. Connecting D to the complementary output makes the first flip-flop toggle, allowing two LEDs to indicate complementary states. A single-sided PCB concept has been outlined; the final CAD layout must be electrically verified before fabrication.

References

1. NE555 Timer IC datasheet, manufacturer documentation.
2. 74HC74 Dual D-Type Flip-Flop datasheet, manufacturer documentation.
3. KiCad Documentation, PCB Editor and Gerber export workflow.